WORKED SOLUTION

STEP I 1994, Question 8

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q8

The key idea

The substitution $\theta = \frac{1}{4}\pi - \phi$ reflects the interval $[0, \frac{1}{4}\pi]$ onto itself, and the addition formula turns $1 + \tan(\frac{1}{4}\pi - \phi)$ into $2/(1 + \tan\phi)$. So the integral reappears on the right-hand side with a minus sign, and we can solve for it rather than integrate anything difficult. The two later integrals are this same integral in disguise: one after $x = \tan\theta$, the other after half-angle formulae.

Solution

The first integral

Call the integral I. Put $\theta = \frac{1}{4}\pi - \phi$, so $\mathrm{d}\theta = -\mathrm{d}\phi$; when $\theta = 0$, $\phi = \frac{1}{4}\pi$, and when $\theta = \frac{1}{4}\pi$, $\phi = 0$. Hence

$$I = \int_{\frac{1}{4}\pi}^{0} \ln(1 + \tan(\tfrac{1}{4}\pi - \phi))\,(-\mathrm{d}\phi) = \int_{0}^{\frac{1}{4}\pi} \ln(1 + \tan(\tfrac{1}{4}\pi - \phi))\,\mathrm{d}\phi.$$

By the addition formula, using $\tan\frac{1}{4}\pi = 1$,

$$1 + \tan(\tfrac{1}{4}\pi - \phi) = 1 + \frac{1 - \tan\phi}{1 + \tan\phi} = \frac{2}{1 + \tan\phi}.$$

(On $[0, \frac{1}{4}\pi]$ we have $\tan\phi \geq 0$, so nothing vanishes and every logarithm is of a positive number.) Therefore

$$I = \int_{0}^{\frac{1}{4}\pi} (\ln 2 - \ln(1 + \tan\phi))\,\mathrm{d}\phi = \tfrac{1}{4}\pi \ln 2 - I,$$

because the last integral is I again, with the dummy variable renamed.

Ask yourself We never found an antiderivative. Why is that allowed?

We don't need one. I is just a number, and the substitution gives an equation $I = \frac{1}{4}\pi \ln 2 - I$ which that number must satisfy. Solving it,

$$2I = \tfrac{1}{4}\pi \ln 2 \quad \Longrightarrow \quad \int_0^{\frac{1}{4}\pi} \ln(1+\tan\theta)\,\mathrm{d}\theta = \tfrac{1}{8}\pi \ln 2,$$

as required.

The integral with $1 + x^2$

The factor $\dfrac{\mathrm{d}x}{1+x^2}$ suggests $x = \tan\theta$, so $\mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta = (1+x^2)\,\mathrm{d}\theta$. The limits $x = 0$ and $x = 1$ become $\theta = 0$ and $\theta = \frac{1}{4}\pi$. Thus

$$\int_0^1 \frac{\ln(1+x)}{1+x^2}\,\mathrm{d}x = \int_0^{\frac{1}{4}\pi} \ln(1+\tan\theta)\,\mathrm{d}\theta = \tfrac{1}{8}\pi \ln 2.$$

The integral with $\sin x$ and $\cos x$

Call it J. The quickest route is symmetry: substituting $x = \frac{1}{2}\pi - u$ swaps sin and cos and maps $[0, \frac{1}{2}\pi]$ to itself, so

$$J = \int_0^{\frac{1}{2}\pi} \ln\left(\frac{1+\cos u}{1+\sin u}\right)\mathrm{d}u = -J,$$

and hence $J = 0$. (Equivalently, $\int_0^{\frac{1}{2}\pi} \ln(1+\sin x)\,\mathrm{d}x$ and $\int_0^{\frac{1}{2}\pi} \ln(1+\cos x)\,\mathrm{d}x$ are equal, so their difference is zero.)

> **Ask yourself** Where does the first result come in?

It is worth seeing how J is linked to I, which is presumably why the question places it here. With $t = \frac{1}{2}x$, the half-angle formulae give

$$1+\cos x = 2\cos^2 t, \qquad 1+\sin x = (\cos t + \sin t)^2,$$

so

$$\frac{1+\sin x}{1+\cos x} = \frac{(\cos t+\sin t)^2}{2\cos^2 t} = \tfrac{1}{2}(1+\tan t)^2.$$

As x runs over $[0, \frac{1}{2}\pi]$, t runs over $[0, \frac{1}{4}\pi]$, and $\mathrm{d}x = 2\,\mathrm{d}t$. Hence

$$J = \int_0^{\frac{1}{4}\pi} (2\ln(1+\tan t) - \ln 2)\,2\,\mathrm{d}t = 4I - \tfrac{1}{2}\pi\ln 2 = \tfrac{1}{2}\pi\ln 2 - \tfrac{1}{2}\pi\ln 2 = 0,$$

which agrees with the symmetry argument. So

$$\int_0^{\frac{1}{2}\pi} \ln\left(\frac{1+\sin x}{1+\cos x}\right)\mathrm{d}x = 0.$$

Checks

Simpson's rule gives 0.2721982613 for both the tan integral and the $1 + x^2$ integral, matching $\frac{1}{8}\pi \ln 2 = 0.2721982613\ldots$; the third integral comes out as 0 to machine precision. A rough check by hand: the integrand $\ln(1 + \tan\theta)$ rises from 0 to $\ln 2$, so I should be roughly half of $\frac{1}{4}\pi \ln 2$. That is exactly the answer.

Common slips

- Forgetting that $\mathrm{d}\theta = -\mathrm{d}\phi$ *and* that the limits swap. The two sign changes cancel; keeping only one gives $I = -\frac{1}{4}\pi \ln 2 - I$ (and a negative answer to a positive integral).
- Writing $\tan(\frac{1}{4}\pi - \phi) = 1 - \tan\phi$, which forgets the denominator in the addition formula.
- Spending a long time on the third integral without noticing the $x \mapsto \frac{1}{2}\pi - x$ symmetry.

Beyond STEP

The trick of substituting a reflection and adding the result to the original, so that $\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a+b-x)\,\mathrm{d}x$, is called the King property in some texts. It also gives the classic $\int_0^{\frac{1}{2}\pi} \ln\sin x\,\mathrm{d}x = -\frac{1}{2}\pi \ln 2$. At university, results like these sit inside the theory of definite integrals with parameters and contour integration, where whole families of integrals are evaluated without finding an antiderivative. Needham's *Visual Complex Analysis* shows how complex methods produce such results, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with these symmetry tricks.

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